

화학수학

열여섯 번째 수업

행렬 / 군론

Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

matrix elements
(행렬 원소)

(m by n matrix)

Matrix Algebra

- Two matrices are **equal** if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix

- **Sum of two matrices:** $\mathbf{C} = \mathbf{A} + \mathbf{B}$ if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- **Product of a scalar c and a matrix $\mathbf{A}$:** $\mathbf{B} = c\mathbf{A}$ iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

Product of Two Matrices

- Matrix product $\mathbf{C} = \mathbf{AB}$ iff

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

where n is the number of columns in $\mathbf{A}$ and the number of rows in $\mathbf{B}$.

$$(m \text{ by } n \text{ matrix}) \times (n \text{ by } p \text{ matrix}) = (m \text{ by } p \text{ matrix})$$

Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$$\mathbf{AB} \neq \mathbf{BA} \text{ (in some cases)}$$

Identity Matrix (항등행렬) **E**

$$\mathbf{EA} = \mathbf{AE} = \mathbf{A}$$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$E_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Zero Matrix (영행렬) $\mathbf{0}$

$$\mathbf{0A} = \mathbf{A0} = \mathbf{0}$$

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$0_{ij} = 0$$

Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^n a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

Gauss-Jordan Elimination

- Construct an **augmented matrix** (첨가행렬)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[\begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array} \right]$$

- Then make the identity matrix

$$\left[\begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array} \right] A^{-1}$$

Transpose of Matrix (전치행렬) $\tilde{\mathbf{A}}$

Rows $\rightarrow$ columns, columns $\rightarrow$ rows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$(\tilde{\mathbf{A}})_{ij} = \tilde{a}_{ij} = a_{ji}$$

Hermitian Conjugate of Matrix $\mathbf{A}^\dagger$

(에르미트 켤레 행렬)

Transpose + complex conjugate of each element

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\mathbf{A}^\dagger = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$$(\mathbf{A}^\dagger)_{ij} = a_{ji}^*$$

Properties of a Square Matrix

- Singular matrix (특이행렬) has no inverse
- Symmetric matrix (대칭행렬): $\mathbf{A} = \tilde{\mathbf{A}}$
- Hermitian matrix (에르미트행렬): $\mathbf{A} = \mathbf{A}^\dagger$
- Orthogonal matrix (직교행렬): $\mathbf{A}^{-1} = \tilde{\mathbf{A}}$
- Unitary matrix (유니타리행렬): $\mathbf{A}^{-1} = \mathbf{A}^\dagger$

Trace of a Square Matrix (대각합)

$$\text{tr } \mathbf{A} = \sum_{i=1}^n a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \text{tr } \mathbf{A} = a_{11} + a_{22} + a_{33}$$

Even when $\mathbf{AB} \neq \mathbf{BA}$,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

오늘의 수업

- Matrices
 - Determinant
 - Matrix eigenvalue problem
- Group theory
 - Groups
 - Point groups

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Example 13.17

Find the value of the determinant

$$\begin{bmatrix} 3 & -17 \\ 1 & 5 \end{bmatrix}.$$

Using the definition of the 2 by 2 determinant:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$\begin{vmatrix} 3 & -17 \\ 1 & 5 \end{vmatrix} = 3 \times 5 - (-17) \times 1 = 32$$

Expanding the Determinant

1. Pick a row or a column (with many zeros)
2. Calculate $\text{each element} \times \text{sign} \times \text{minor}$
 - **Minor**: determinant after deleting the row and the column containing the element
 - **Sign**: counting the number of steps of one row or one column required to get from the upper left element
→ even number: (+), odd number: (-)
3. Sum them up across the row or the column
4. Repeat this up to a sum of 2 by 2 determinants



Example 13.18

Expand the 3 by 3 determinant of a general matrix $\mathbf{A}$.

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Pick the first row:

$$\det \mathbf{A} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Now we can apply the definition of the 2 by 2 determinant:

$$\det \mathbf{A} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Properties of the Determinant (1)

- If every element in any one row or in any one column is zero, the value of the determinant is zero.
- If two rows or two columns are identical, the determinant has value zero.
- If two rows or two columns are interchanged, the determinant = $(-1) \times (\text{original determinant})$.

Properties of the Determinant (2)

- If every element in any one row or in any one column is multiplied by same scalar c , the value of the determinant = $c \times$ (original determinant).
- Row operations (or column operations) do not change the value of the determinant.

$$\begin{vmatrix} a_{11} + ca_{21} & a_{12} + ca_{22} & a_{13} + ca_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Properties of the Determinant (3)

- The determinant of a triangular matrix is equal to the product of the diagonal elements.

$$\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33}$$

- The determinant of a matrix and that of its transpose are equal.

$$\det \tilde{\mathbf{A}} = \det \mathbf{A}$$

Determinant in Quantum Mechanics

An n -electron wave function $\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n)$ must obey

1. Antisymmetric condition:

$$\Psi(\vec{r}_2, \vec{r}_1, \dots, \vec{r}_n) = -\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n)$$

2. Pauli exclusion principle:

$$\Psi(\vec{r}_1, \vec{r}_1, \dots, \vec{r}_n) = 0$$

Hartree Approximation

- Multi-electron wave function $\leftarrow$ one-electron wave function (independent electron assumption)

$$\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n) = \psi_1(\vec{r}_1)\psi_2(\vec{r}_2) \cdots \psi_n(\vec{r}_n)$$

- The Hartree wave function does not meet the requirements (antisymmetric condition, exclusion principle)

Hartree-Fock Approximation

- Slater determinant: resolves the problem

$$\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n) = \frac{1}{\sqrt{n!}} \begin{vmatrix} \psi_1(\vec{r}_1) & \psi_1(\vec{r}_2) & \cdots & \psi_1(\vec{r}_n) \\ \psi_2(\vec{r}_1) & \psi_2(\vec{r}_2) & \cdots & \psi_2(\vec{r}_n) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_n(\vec{r}_1) & \psi_n(\vec{r}_2) & \cdots & \psi_n(\vec{r}_n) \end{vmatrix}$$

- Antisymmetric condition:

$$\Psi(\vec{r}_2, \vec{r}_1, \dots, \vec{r}_n) = -\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n)$$

- Pauli exclusion principle:

$$\Psi(\vec{r}_1, \vec{r}_1, \dots, \vec{r}_n) = 0$$

Other Applications of the Determinant

- Vector product of two vectors
- Determinant of a singular matrix
- Matrix eigenvalue problem

Vector Product of Two Vectors

The vector product (cross product) of $\vec{A}$ and $\vec{B}$ is

$$\begin{aligned}\vec{A} \times \vec{B} &= \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_z B_x - A_x B_z) + \hat{k}(A_x B_y - A_y B_x) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}\end{aligned}$$

Determinant of a Singular Matrix

- A matrix with no inverse is **singular**
- The determinant of a singular matrix has value 0

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1/2 & 1 \end{bmatrix}$$

$$\det \mathbf{A} = 1 \times 1 - 2 \times (1/2) = 0$$

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

We can rearrange it as

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

Use of Determinant

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

If $\mathbf{B} - b\mathbf{E} \neq \mathbf{0}$ and $\vec{x} \neq \vec{0}$, we have

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

By solving this equation, one can determine b and consequently $\vec{x}$.

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

Using the determinant condition, $\det(\mathbf{B} - b\mathbf{E}) = 0$,

$$\begin{vmatrix} 1-b & 1 & 0 \\ 1 & 1-b & 1 \\ 0 & 1 & 1-b \end{vmatrix} = 0$$

Calculate the determinant:

$$(1-b) \begin{vmatrix} 1-b & 1 \\ 1 & 1-b \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 0 & 1-b \end{vmatrix} = (1-b)[(1-b)^2 - 1] - (1-b) = 0$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

$$(1 - b)[(1 - b)^2 - 2] = 0$$

Solving this equation, we obtain the eigenvalues

$$b = 1 \text{ or } b = 1 \pm \sqrt{2}$$

Now determine the eigenvectors by letting

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

1) $b = 1$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = x_1 \\ x_1 + x_2 + x_3 = x_2 \\ x_2 + x_3 = x_3 \end{cases}$$

Hence, $x_2 = 0$ and $x_3 = -x_1$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

1) $b = 1$: $x_2 = 0$ and $x_3 = -x_1$

The eigenvector corresponding to $b = 1$ is

$$\vec{x}(b = 1) = \begin{bmatrix} x_1 \\ 0 \\ -x_1 \end{bmatrix} = C \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

(C : arbitrary constant)

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

2) $b = 1 + \sqrt{2}$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = (1 + \sqrt{2}) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = (1 + \sqrt{2})x_1 \\ x_1 + x_2 + x_3 = (1 + \sqrt{2})x_2 \\ x_2 + x_3 = (1 + \sqrt{2})x_3 \end{cases}$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

2) $b = 1 + \sqrt{2}$:

$$\begin{cases} x_2 = \sqrt{2}x_1 \\ x_1 + x_3 = \sqrt{2}x_2 \\ x_2 = \sqrt{2}x_3 \end{cases}$$

Hence, $x_2 = \sqrt{2}x_1$ and $x_3 = x_1$.

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

2) $b = 1 + \sqrt{2}$: $x_2 = \sqrt{2}x_1$ and $x_3 = x_1$

The eigenvector corresponding to $b = 1 + \sqrt{2}$ is

$$\vec{x}(b = 1 + \sqrt{2}) = \begin{bmatrix} x_1 \\ \sqrt{2}x_1 \\ x_1 \end{bmatrix} = C \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}$$

(C : arbitrary constant)

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

3) $b = 1 - \sqrt{2}$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = (1 - \sqrt{2}) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = (1 - \sqrt{2})x_1 \\ x_1 + x_2 + x_3 = (1 - \sqrt{2})x_2 \\ x_2 + x_3 = (1 - \sqrt{2})x_3 \end{cases}$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

3) $b = 1 - \sqrt{2}$:

$$\begin{cases} x_2 = -\sqrt{2}x_1 \\ x_1 + x_3 = -\sqrt{2}x_2 \\ x_2 = -\sqrt{2}x_3 \end{cases}$$

Hence, $x_2 = -\sqrt{2}x_1$ and $x_3 = x_1$.

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

3) $b = 1 - \sqrt{2}$: $x_2 = -\sqrt{2}x_1$ and $x_3 = x_1$

The eigenvector corresponding to $b = 1 - \sqrt{2}$ is

$$\vec{x}(b = 1 - \sqrt{2}) = \begin{bmatrix} x_1 \\ -\sqrt{2}x_1 \\ x_1 \end{bmatrix} = C \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$$

(C : arbitrary constant)

Group (군)

Collection of members
with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the **product** AB is also a member of the group
2. The group must contain the **identity** E such that

$$AE = EA = A \text{ (for every member } A\text{)}$$

3. The **inverse** of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

4. The **associative law** holds: $A(BC) = (AB)C$



Example

group of $\{1, i, -1, -i\}$

- Multiplication table

	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

Identity = 1